

The Litehaus Protocol: Spatio-Temporal Cryptography via Vigesimal Tallying and Cellular Automata

Abstract

This paper outlines the architecture of the “Litehaus” beacon, a decentralized time-locked puzzle system. By integrating the Mayan Long Count (vigesimal) calendar with Sylvester–Hadamard matrices and 2D cellular automata (CA), we define a “Proof of Solve” mechanism that transitions successful participants from passive observers (Blue Team) to active network adversaries (Red Team).

1 Temporal Anchoring: The Long Count Epoch

The Litehaus utilizes a linear tally of days, modeled after the Mayan Long Count, to establish a drift-free temporal baseline. Unlike the Gregorian leap-year system, the Long Count is a base-20 (vigesimal) counter. The target epoch for the “Saboteur’s Ascension” is defined as:

$$13.0.13.9.0 \equiv \text{May 20, 2026.}$$

The precision of the signal is maintained via Stratum-1 synchronization, where the Pulse Per Second (PPS) acts as the logic gate for the second phase of the puzzle.

2 Spatial Mapping: The Sylvester–Hadamard Core

The broadcast packet consists of a sparse JSON array representing coordinates in a 32×32 grid. The underlying structure is a Hadamard matrix H_n , defined such that:

$$H_n H_n^T = n I_n,$$

where I_n is the identity matrix. To generate the 32×32 grid, we use the recursive Sylvester construction:

$$H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}.$$

Participants must solve for missing values ($v \in \{1, -1\}$) to complete the matrix, effectively performing a high-order parity check on the received signal.

3 Temporal Evolution: Cellular Automaton Logic

A solved Hadamard matrix remains a “dead” state until the Temporal Trigger is applied. At $t_{\text{epoch}} + 1$ s, the grid must be iterated through a single step of a 2D cellular automaton, specifically the Moore-neighborhood rule B3/S23 (Conway’s Life):

$$\sigma_{i,j}^{t+1} = f\left(\sigma_{i,j}^t, \sum \text{neighbors}\right).$$

This iteration collapses the mathematical fractal into a high-density information structure—a QR-compliant pattern containing the “Vault Access Token.”

4 Adversarial Transition: Red Team Mechanics

Upon successful resolution, the “Blue Team” solver is granted access to the Red Team Spoofing Toolkit. The mathematical challenge shifts from interpretation to interference.

4.1 Temporal Spoofing

Red Team actors utilize NTP-offsetting to broadcast a false t_{epoch} . If the target node synchronizes to a spoofed pulse:

$$\Delta t \neq 0 \implies \text{CA Evolution Failure.}$$

The resulting grid will exhibit “high-entropy noise” rather than a valid QR pattern.

4.2 Sub-Pixel Jitter and Edge Sabotage

By exploiting the difference between Android (ZXing) and iPhone (VisionKit) sampling algorithms, Red Teamers can inject sub-pixel variances into the grid’s Quiet Zone. This increases the “Error Correction” overhead (ECC) beyond the decodable threshold for specific hardware, effectively “blacking out” certain segments of the participant pool.

5 Conclusion: The Self-Hardening Network

The Litehaus Protocol creates a recursive loop of network hardening. Each successful solve increases the number of potential adversaries, which in turn necessitates higher signal-to-noise ratios and more robust hardware for the next wave of participants. It is a digital ecosystem that thrives on its own resistance.